

INTERACTIVE RECREATION OF JOHN SNOW'S 1854 LONDON CHOLERA EPIDEMIC

ABOUT THE TASK

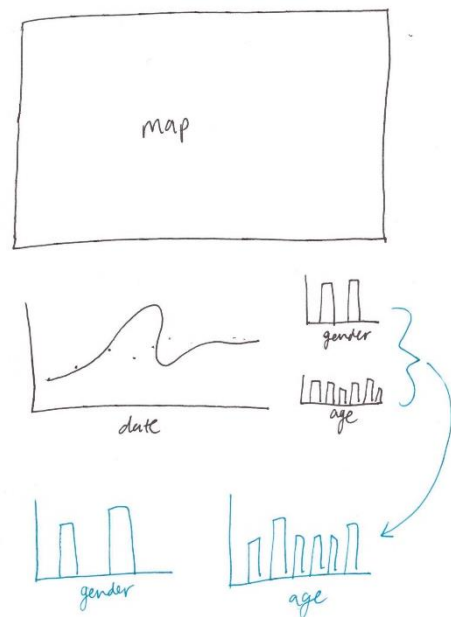
Dr. John Snow, the father of modern epidemiology, is famous for the map he created in 1854 of the cholera epidemic in London. His map was used to link the outbreak of cholera to contaminated water – specifically at the Broad Street Pump. For this project, I was tasked with creating an interactive version of Snow's map in D3.js. I used the following data sets:

- Street data – file contained the X and Y coordinates of the streets
- Pump data – file contained the X and Y coordinates of the pumps
- Death data – file contained the number of deaths for each of the 42 days
- People data – file contained the X and Y coordinates that they person died as well as gender and age category

DESIGN PROCESS

ITERATION 1

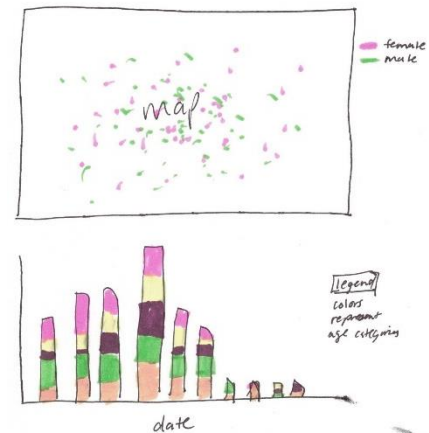
The first code I wrote plotted the pumps since there were only 13 and I had an example from class to create a scatter plot. From there, I added the 571 people using the same scatter plot method and finally drew the streets in a similar method as a line graph. I also created static plots of the death distribution by date, gender and age. At the beginning, I really struggled to maintain consistency in my layout because things would wildly change depending on my screen size. I decided I need to simplify and add feature to connect the demographic data to the map.



In reality, the plots ended up underneath the date chart. I had a lot of instability in the location of plots. User couldn't view all objects at once. Plan: simplify to less visuals

ITERATION 2

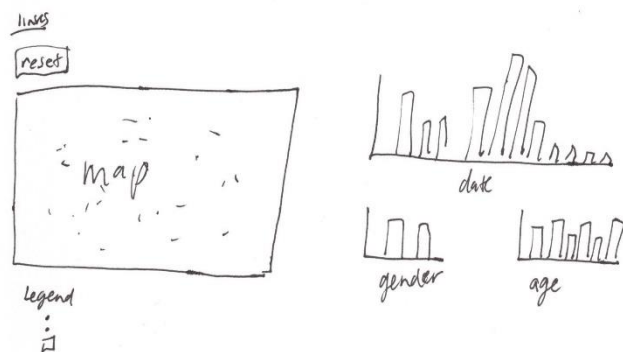
In this second iteration, I colored the points based on the sex of the person. I considered downsizing to one graph that showed the age distribution for each day while the map showed the sex. I decided against this because there are six age categories and the human brain is not very accurate in estimating values in stacked bar charts.



This condensed my data but made it busy and the bar chart can be deceiving with the age distribution. In this iteration, map is always color coded by gender.

ITERATION 3

In my final interaction, I focused on readability, simplicity, and interactivity. I limited my colors and used them sparingly only when color would add meaning. I finally figured out how to anchor my svg's so they stayed put no matter what the screen size was. On most laptop screens, all elements should be visible without scrolling. I also added the ability to click on any bar of the three charts and highlight the data on the map linked to the bar. Data for each element can be viewed in a tooltip. I chose against displaying the info on the chart to keep the view clean and ease of viewing the whole distribution. If a user wants to see specific values for a bar, a tooltip pops up on a simple mouse over. The user can also zoom and pan on the map and a reset button resets the map and all charts.



I finally anchored the svgs so they didn't float around when the page changed. Now I could add labels, legends and fix the layout to be pleasing and simple.

FUTURE GOALS

With more time, my next step would be to add the ability to click on multiple days on the date chart to display on the map. My idea was to create a data set that held all the information for each click and filter the chart on that data set instead of one single click. The extended data set would be cleared with the reset button.

DESIGN RATIONALE

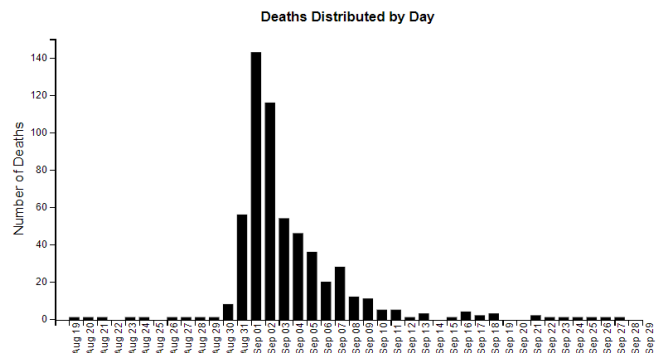
I opted to keep the design simple with the layout and especially colors. I used white or black as much as possible and only used color if it would add meaningful information. The streets are light blue because the street data is important, but it should not overshadow the points. I kept the background white for the most contrast against the street, pump, and people data. Adding color to the bar charts would not add information so I kept them black. Elements do change to yellow when they are highlighted by the mouse. I checked my visuals with ColorOracle and found there are no conflicting colors in the different views. Users can zoom in on the map and pan to view the data up close.

Reset

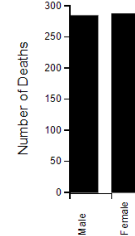


Legend

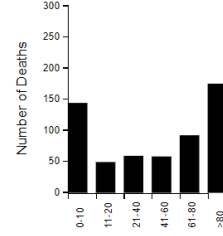
- Death
- Water Pump
- Building



Deaths Distributed by Gender

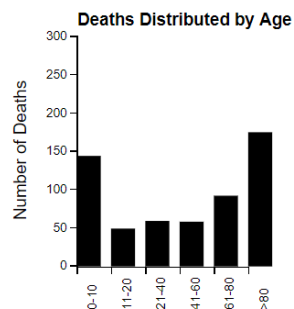


Deaths Distributed by Age



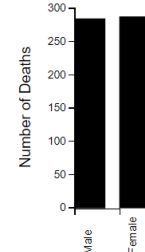
DISCOVERY

I was curious if the distribution of death would be similar for males and females since females likely stayed at home all day while the males could have gotten water at a different site while away at work for the day. However, the number of deaths of females and males is comparable for all days of death.



I was not surprised to see that the youngest and oldest age groups had the most deaths. These are generally the two most vulnerable age groups to disease.

Deaths Distributed by Gender



VIDEO TUTORIAL

In this video (<https://www.youtube.com/watch?v=rxCpPn1I-z0>), I demonstrate how to

- Zoom in and out of the map and pan on the map
- Use the tooltip to find out more information
- Click on a bar in the date distribution graph to update the gender and age distribution graphs and highlight those points on the map
- View the map by gender
- View the map by age

REFERENCES

I based my map on the scatter plot and line plot examples from class (IUPUI H517, Professor Khairi Reda). I also integrated the linking example. Other references include:

Zoom feature: <http://bl.ocks.org/shawnbot/6518285>

Zoom reset: <https://bl.ocks.org/catherinekerr/b3227f16cebc8dd8beee461a945fb323>

Tooltips: <http://bl.ocks.org/d3noob/a22c42db65eb00d4e369>

D3.js book: <https://www.oreilly.com/library/view/interactive-data-visualization/9781449340223/>